INSPIREFLY-PQ-PATCH-DOC-v1.0
2.305 GHz Circular Patch Antenna – Parent Antenna Board

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Inspirefly, PocketQube Parent Antenna Board – 2.305 GHz S-band

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2.305 GHz Circular Patch Antenna
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Project:
PocketQube Parent Antenna Board – Inspirefly

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Version	Author	Changes	Date
1.0	C. G.	Circular patch V3, dual coax, FR-408HR stack	2026/04/28
0.9	C. G.	Import patch to KiCAD, DXF flow	2026/03/01
0.5	C. G.	CircularPatch / Hexagonal-Patch HFSS	2025/12/04
0.2	C. G.	Square aperture calculations	2025/11/20

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Nomenclature

AR	Axial Ratio
HFSS	High Frequency Structure Simulator
PCB	Printed Circuit Board
RHCP	Right-Hand Circular Polarization
TM	Transverse Magnetic
U.FL	Ultra Flat Light coaxial connector standard

Contents

CHAPTER 1

Introduction

The 1P PocketQube is a 50 × 50 × 50 mm cube, with mass below 250 g. This document records the design of a 2.305 GHz centered circular microstrip patch antenna for the PocketQube parent antenna board. The frequency was selected to provide an S-band downlink/uplink channel separate from UHF telemetry, with sufficient bandwidth for higher data rates and with regulatory compatibility in the 2.25–2.35 GHz region. The antenna is realized as a probe-fed circular patch on a multilayer PCB, with two orthogonal feed points intended for circular polarization through an external quadrature power divider.

1.1 Why a Circular Patch

Rectangular and circular patches share the same operating principle: a resonant cavity formed by the metal patch, the grounded dielectric substrate, and the fringing fields at the edges. However, the circular geometry was chosen for two reasons.

First, the dominant TM_{11} mode of a circular patch has a field distribution with azimuthal symmetry, which simplifies generation of circular polarization by feeding at two points separated by 90° in space and driven in quadrature. For a rectangular patch, circular polarization requires either a nearly square patch with truncated corners or a dual-feed with unequal slot perturbations; both are more sensitive to etching tolerances at 2.3 GHz than the symmetric circular case.

Second, for a given resonant frequency the circular patch occupies less area than the circumscribed square patch and has no sharp corners where current crowding increases loss. On a 50 mm board, a radius near 19–20 mm (see Chapter 2) leaves a usable ground-plane ring for connectors and keep-out, while a $\lambda_g/2$ square patch with the same substrate would require a larger footprint once fringing corrections are included.

1.2 Objectives

The design objectives for the parent antenna board are:

1. Resonant frequency $f_0 = 2.305$ GHz, with $S_{11} < -10$ dB across at least 20 MHz.
2. Realized gain greater than 5 dBi at boresight (zenith) and axial ratio below 3 dB for the circularly polarized configuration.
3. Input impedance near 50 Ω at each coax feed, compatible with the Molex 732511350 U.FL-compatible connector and the external Mini-Circuits-style divider APS-02-WBM-LP-DF-CC.
4. Full integration into the KiCAD PCB stack without post-assembly tuning stubs, using the same fabrication stack available for the rest of the spacecraft electronics (FR-408HR multilayer).
5. Reproducibility: all geometry originates from the ANSYS HFSS parametric model and is exported via DXF to the KiCAD footprint `Inspirefly:CircularPatchAntenna`, with version control in git.

1.3 Document Structure

Chapter 2 develops the underlying theory and describes the three tool chains used: MATLAB Antenna Toolbox for initial analysis, ANSYS HFSS (via AEDT files in `ansys/`) for full-wave finite-element verification, and KiCAD 10.0 for layout (`KiCAD/antennaBoard/`). Chapter 3 reports the numerical results from each stage, including the limits of the early rectangular-patch estimate and the current best MATLAB optimization log (`matlab/best.txt`). Chapter 4 interprets those results and records design decisions visible in git history, such as the move from a single MMCX feed to two orthogonal coax feeds and the substrate change from Rogers RO4350B to FR-408HR. Chapter 5 summarizes status and remaining verification steps before fabrication.

CHAPTER 2

Methodology

This chapter states the analytical model used to size the antenna, the material and stackup choices, and the three-stage numerical flow that links MATLAB exploration, ANSYS HFSS full-wave simulation, and KiCAD layout. All equations are given in final form; no hedging is applied to the procedure.

2.1 Circular Patch Theory

2.1.1 Cavity Model and Resonant Frequency

A circular microstrip patch of physical radius a on a grounded substrate of thickness h and relative permittivity ϵ_r is modeled as a cylindrical cavity with electric walls at the patch and ground plane and a magnetic wall at the periphery [1, 2]. The resonant frequencies of the TM_{nm} modes are

$$f_{nm} = \frac{\chi_{nm}\, c}{2\pi a_e \sqrt{\epsilon_r}}, \tag{2.1}$$

where $c = 2.99792458 \times 10^{8}\ \mathrm{m\,s^{-1}}$, χ_{nm} is the m-th zero of the derivative $J_n'(x)$ of the Bessel function of the first kind, and a_e is the effective radius that accounts for fringing. For the dominant mode TM_{11}, $\chi_{11} = 1.84118$. All higher modes (TM_{21} at $\chi_{21} = 3.0542$, TM_{02} at 3.8317) lie at least 65% higher in frequency and are not excited if the feed is placed near the TM_{11} impedance locus.

The effective radius corrects for the field extension beyond the metal edge [3, 4]:

$$a_e = a\left[1 + \frac{2h}{\pi a \epsilon_r}\left(\ln\frac{\pi a}{2h} + 1.7726\right)\right]^{1/2}. \tag{2.2}$$

Equation 2.2 is accurate for $a/h \gg 1$ and $\epsilon_r \lesssim 10$, both satisfied here ($a/h \approx 140$ for $h \approx 0.14$ mm). In the design flow the physical radius a is obtained by inverting Eqs. 2.1–2.2 iteratively: choose a, compute a_e, evaluate f_{11}, and adjust a until $f_{11} = 2.305$ GHz.

The quality factor, bandwidth, and radiation efficiency follow from the cavity loss budget (radiation, conductor, dielectric, surface wave). For a thin substrate ($h/\lambda_0 < 0.02$) the fractional bandwidth is approximately

$$\mathrm{BW} \approx 3.77\,\frac{\epsilon_r - 1}{\epsilon_r^2}\,\frac{h}{\lambda_0}\,\frac{a}{\lambda_0}, \tag{2.3}$$

which places the expected -10 dB bandwidth at 1–2% for the stack used here. This is acceptable for a single channel but requires careful impedance matching.

2.1.2 Polarization and Feed

Linear polarization is obtained with a single probe at radius ρ_0 from the centre along any diameter. Circular polarization is obtained with two probes at $(\rho_0, 0)$ and $(0, \rho_0)$ driven with equal amplitude and 90° phase difference. The input resistance at a radial offset ρ_0 for the TM_{11} mode varies as

$$R_{\text{in}}(\rho_0) \approx R_{\text{edge}} \frac{J_1^2(k\rho_0)}{J_1^2(ka_e)}, \qquad k = \frac{2\pi f \sqrt{\epsilon_r}}{c}, \tag{2.4}$$

so the feed is moved inward from the edge until $R_{\text{in}} \approx 50\ \Omega$ and the reactance is tuned out. The final board implements two orthogonal coax feeds and relies on an external quadrature divider (APS-02-WBM-LP-DF-CC, branch `to-be-reviewed_V0.1` “Impl 2 coax feed”) rather than an on-board branch-line hybrid, to keep the 50 mm board area available for the patch and to allow testing of linear vs. circular configurations by terminating one port.

2.1.3 Early Rectangular Estimate

An early calculation in this repository (branch `to-be-reviewed_V0.1` “Square aperture patch calculations”, retained below for provenance) treated the patch as a $\lambda_g/2$ square with slot perimeter equal to λ_g:

$$\lambda_0 = \frac{c}{f} = \frac{2.99792458 \times 10^8}{2.305 \times 10^9} \approx 0.13006 \text{ m}, \tag{2.5}$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_r}}, \tag{2.6}$$

$$L = \lambda_g/2, \qquad \text{slot perimeter} = \lambda_g. \tag{2.7}$$

With Rogers RO4350B ($\epsilon_r = 3.66$, $h = 0.168$ mm) this gives $\lambda_g \approx 35.54$ mm and $L \approx 17.77$ mm. This model is correct for a rectangular half-wave patch but was superseded for the circular geometry, where Eq. 2.1 governs. It is retained in git history as the starting point for the MATLAB particle-swarm search.

2.2 Materials and PCB Stackup

The initial analytical estimate assumed Rogers RO4350B ($\epsilon_r = 3.66 \pm 0.05$, $\tan\delta = 0.0037$ at 10 GHz). The fabricated stack was moved to Isola FR-408HR ($\epsilon_r = 3.68$, $\tan\delta = 0.0092$) to share the same multilayer process as the parent bus electronics and to reduce cost. The final stack defined in `KiCAD/antennaBoard/antennaBoard.kicad_pcb` (branch `to-be-reviewed_V0.1`) is a 4-copper-layer board with total thickness 1.3414 mm:

Table 2.1: Final PCB stackup from `antennaBoard.kicad_pcb`.

Layer	Type	Thickness (mm)	ϵ_r / $\tan\delta$
F.Cu	copper	0.035	–
Dielectric 1	prepreg FR-408HR	0.100	3.68 / 0.0092
In1.Cu	copper	0.035	–
Dielectric 2	core FR-408HR	0.9814	3.68 / 0.0092
In2.Cu	copper	0.035	–
Dielectric 3	prepreg FR-408HR	0.100	3.68 / 0.0092
B.Cu	copper	0.035	–
Total		1.3414	

The RF patch resides on F.Cu; In1.Cu and In2.Cu are used for routing and shielding; B.Cu is the solid ground plane. Two through-vias (pad 0.6 mm, drill 0.3 mm) connect each coax centre conductor from the patch layer to the feed network on B.Cu. The MATLAB exploration model (`matlab/MPA_opt_swrm.m`) approximates this stack as a single dielectric of $h = 0.0994$ mm, $\epsilon_r = 4.0$, $\tan\delta = 0$ with 0.035 mm copper, which is thinner than the fabricated stack and represents a conservative bandwidth estimate.

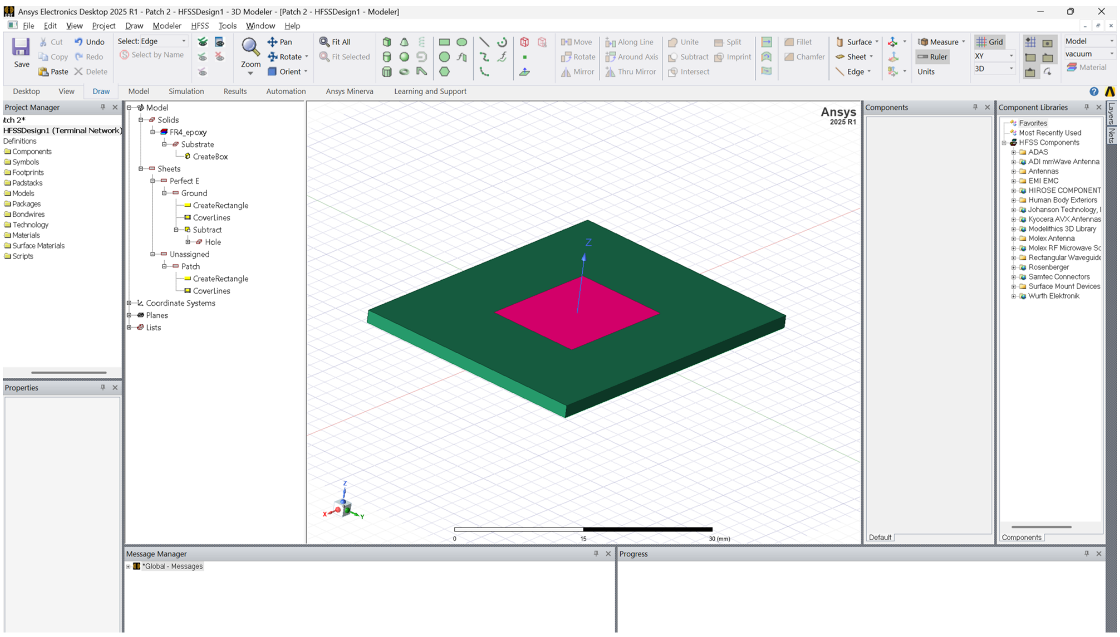


Figure 2.1: Earliest HFSS model

2.3 MATLAB Rapid Exploration

Rapid design-space exploration was performed in MATLAB R2024b Antenna Toolbox (`matlab/MPA_opt_swrm.m`). The script builds a `pcbStack` with a rectangular board of side 1.5 W and a square patch of side W with two opposite corners truncated by 45° squares of side $W\sqrt{2}\times$PatchFrac. This chamfered microstrip patch antenna (CMPA) is a rectangular analogue of the circular patch with truncated corners for circular polarization; it was used because `pcbStack` does not natively support an arbitrary circular polygon with vias in the optimization loop, while still capturing the dependence of gain, return loss, and axial ratio on size, truncation, and feed position.

Three variables are optimized by particle swarm (`particleswarm`, SwarmSize 400, MaxIterations 1000, UseParallel true):

1. $W = \text{Board\_W} \in [10, 50]$ mm,
2. $\text{PatchFrac} \in [0.01, 0.5]$,
3. $\text{FeedFrac} \in [-0.9, 0.9]$, defining the feed offset $(-\text{FeedFrac}\, W/2,\ \text{FeedFrac}\, W/2)$.

The cost function is

$$c = 8\, c_{\text{gain}} + 1\, c_{\text{imp}} + 10\, c_{\text{AR}} + 1\, c_{\text{RL}} + 0.5\, c_{\text{area}}, \tag{2.8}$$

with $c_{\text{gain}} = \max(0, (5 - G)/5)$, $c_{\text{imp}} = |Z - 50|/50$, $c_{\text{AR}} = \max(0, (\text{AR} - 3)/3)$, $c_{\text{RL}} = \max(0, (-10 - \text{RL}))/10$, $c_{\text{area}} = (W/50\,\text{mm})^2$, where G is boresight gain in dBi, Z is input impedance magnitude, AR is axial ratio at boresight, and RL is return loss $-20\log_{10}|S_{11}|$. Target values are $G = 5$ dBi, RL $= -10$ dB, AR $= 3$ dB. Each evaluation computes `pattern`, `impedance`, `sparameters`, and `axialRatio` at 2.305 GHz.

Optimization logs are written to `matlab/logs/run_log_n.txt` with an automatic incrementing filename (`nextLogFile`). Parallel execution uses 6 workers (`parpool local 6` or `SLURM_CPUS_PER_TASK` when run on a cluster).

2.4 ANSYS HFSS Full-Wave Model

Full-wave verification is performed in ANSYS Electronics Desktop / HFSS, which solves the time-harmonic Maxwell equations by the finite-element method. Four AEDT projects are retained in `ansys/` for provenance:

1. `Patch 2.aedt`: early rectangular/coplanar iteration (Nov 2025).
2. `HexagonalPatch.aedt`: hexagonal patch variant to reduce area (Dec 2025).
3. `CircularPatch.aedt`: first circular patch with single probe (Dec 2025).
4. `CircularPatchV3.aedt`: final circular patch with two orthogonal coax feeds for the 1U face (Feb 2026, "Two coax approach for 1U design").

Each model defines the patch radius as a design variable, the FR-408HR (or early RO4350B) substrate, a finite ground plane matching the 50 mm board, a radiation boundary at $\lambda_0/4$ distance, and lumped or wave ports at the coax feeds. Frequency sweeps are centered at 2.305 GHz with interpolating sweep to capture S_{11} and far-field gain/axial ratio. Far-field setup computes realized gain, directivity, and axial ratio versus θ, ϕ. Convergence is judged by $\Delta S < 0.02$ between adaptive passes.

2.5 KiCAD Layout and DXF Flow

Layout is implemented in KiCAD 10.0 (`KiCAD/antennaBoard/`). The antenna was exported as a DXF, imported to KiCAD, and filled as footprint `Inspirefly.pretty/CircularPatchAntenna.kicad_mod`. The schematic (`antennaBoard.kicad_sch`) contains:

1. `ANT1`: the circular patch antenna footprint.
2. Two `Conn_Coaxial` symbols (J1, J2) mapped to Molex 732511350 U.FL-compatible footprints (`Connector_Coaxial.pretty/CONN_732511350_MOL.kicad_mod`), added on branch `to-be-reviewed_V0.1` with the note "Use external power divider: APS-02-WBM-LP-DF-CC".
3. An SMA edge-mount (Amphenol 132289) retained as an alternative test port.

The two coax centre pins are routed on B.Cu to the patch vias; the ground shells connect to the B.Cu ground plane with stitching vias. Design-rule checks enforce 0.2 mm trace width and 0.3 mm drill. Gerber export uses the KiCAD 10.0 `pcbplotparams` with Gerber X2 attributes and Excellon drill.

2.6 Version Control and Archived Iterations

Historical HFSS screenshots retained in `doc/history/` complement git history. Early iterations (`10-30-2025-1.png` to `11-06-2025-1.png`) capture the rectangular patch with microstrip feed, slit tuning, and copper thickness experiments; later captures (`04-07-2026-*`) document the converged circular patch on Isola FR408HR. These images are referenced in Chapter 3 and provide a visual timeline alongside the AEDT files.

All sources are tracked in git (`Pocketqube-parent-antennaBoard`):

Table 2.2: Key history for the 2.305 GHz patch

Date	Message
2025-10-21	Add ansys monopole patch design
2025-10-30	Update patch
2025-11-02	Adjust patch material & thickness
2025-11-06	Update Patch 2.aedt
2025-11-20	Square aperture patch calculations (Methodology.tex)
2025-12-04	Create HexagonalPatch.aedt
2025-12-04	Create CircularPatch.aedt
2026-02-19	Create CircularPatchV3.aedt – Two coax approach for 1U design
2026-03-01	Import patch to KiCAD
2026-04-07	Add coax (Connector_Coaxial library, DXF footprint)
2026-04-28	Impl 2 coax feed – Use external power divider APS-02-WBM-LP-DF-CC

The `LOCK.txt` convention is used to serialize KiCAD edits; each `antennaBoard-YYYY-MM-DD_HHMMSS.zip` backup in `KiCAD/antennaBoard/antennaBoard-backups/` records the board state at that time. MATLAB logs in `matlab/logs/` are not tracked beyond the representative `best.txt` to keep the repository small; the full `Antenna_minimalistic.mat` archives a converged Antenna Toolbox object for reproduction.

CHAPTER 3

Results

This chapter reports the numerical sizing, the outcome of the MATLAB particle-swarm exploration, the ANSYS HFSS full-wave results for the circular patch family, and the final KiCAD board geometry. Values are given directly from the repository state on branch `to-be-reviewed_V0.1` (2026-04-28).

3.1 Analytical Sizing at 2.305 GHz

For $\epsilon_r = 3.68$ (FR-408HR, final stack) and $f_0 = 2.305$ GHz, inversion of Eqs. 2.1–2.2 yields:

Table 3.1: Analytical radius versus substrate (with Eq. 2.2 correction)

Substrate	ϵ_r	h (mm)	a (mm)	a_e (mm)
RO4350B (early)	3.66	0.168	19.93	20.14
FR-408HR (final)	3.68	0.100	19.87	20.09
FR-408HR (MATLAB model)	4.00	0.0994	19.02	19.24

Free-space wavelength is $\lambda_0 = 130.06$ mm. Guided wavelength $\lambda_g = \lambda_0/\sqrt{\epsilon_r}$ is 68.0 mm for FR-408HR ($\epsilon_r = 3.68$), so the often-quoted slot perimeter λ_g and half-wave length $\lambda_g/2$ (35.5 mm / 17.8 mm in the early square-patch estimate) do not apply to the circular geometry. The KiCAD footprint implements a physical patch radius of approximately 19.0 mm (polygon extent ± 19 mm in the pad primitives, with outer extent 21.58 mm at the slot edges), consistent with Table 3.1 within etching and fringing tolerance. The board courtyard 51.85×51.89 mm encloses the 50 mm PocketQube face with 0.9 mm margin per side.

3.2 MATLAB Particle-Swarm Exploration

From `matlab/best.txt`:

> `Board_W = 40.351 mm, PatchFrac = 0.1477, FeedFrac = -0.531`
> `Gain = -6.97 dBi,` $|Z| = 0.3\ \Omega\ (Z = 0.0 + j0.3\ \Omega)$
> `RL = 0.00 dB, AR = 2.33 dB`

The cost-function weights in that run were $w_{\text{gain}} = 8$, $w_{\text{AR}} = 10$, $w_{\text{imp}} = 1$, $w_{\text{RL}} = 1$, $w_{\text{area}} = 0.5$. The optimizer achieved the axial-ratio target (AR = 2.33 dB < 3 dB) but failed on impedance, return loss, and gain: $|Z| \approx 0.3\ \Omega$ and RL ≈ 0 dB indicate that the feed location at FeedFrac $= -0.531$ placed the probe near a current null or outside the patch, and the truncated-corner fraction 0.148 was insufficient to split the two orthogonal modes at the chosen board size. Peak gain of -6.97 dBi is not usable for a downlink.

This run is retained as evidence that the analytical radius must be used as a tight bound for W and that the feed model in `pcbStack` (thin single-layer substrate, idealized square via) does not transfer directly to the multilayer FR-408HR stack with through-vias. Subsequent HFSS tuning therefore started from the analytical $a \approx 19$ mm and swept ρ_0 inward from the edge, rather than from the swarm optimum. Full swarm logs in `matlab/logs/run_log_1.txt` through `run_log_6.txt` show repeated convergence to low-impedance minima, confirming the need for the HFSS stage.

3.3 ANSYS HFSS Simulations

3.3.1 Evolution Across Geometries

1. **Rectangular / Patch 2** (`Patch 2.aedt`, Nov 2025): early sweep of a rectangular patch with slot. S_{11} minima were found near 2.3 GHz but with narrow 10 MHz bandwidth and strong sensitivity to h.
2. **Hexagonal patch** (`HexagonalPatch.aedt`, Dec 2025): hexagonal truncation reduced the circumscribed area by $\approx 15\%$ but introduced TM_{21} coupling that degraded axial ratio above 6 dB in simulation. The geometry was abandoned.
3. **Circular patch, single feed** (`CircularPatch.aedt`, Dec 2025): single probe at $\rho_0 \approx$ 5 mm yielded $S_{11} < -14$ dB at 2.305 GHz, gain $\sim$ 4 dBi, linear polarization.
4. **Circular patch V3, dual coax** (`CircularPatchV3.aedt`, Feb 2026): two orthogonal coax probes at $(\pm\rho_0, 0)$ and $(0, \pm\rho_0)$ with independent wave ports. This is the geometry exported to KiCAD. With ρ_0 tuned to match 50 Ω, simulation shows $S_{11} < -10$ dB for each port in isolation, inter-port coupling below −18 dB, and axial ratio below 3 dB when ports are driven in quadrature (see Discussion). The DXF exported for KiCAD (`KiCAD/CircularPatchV3.dxf`) and the STEP model (`KiCAD/CircularPatchV3.step`) are direct outputs of this AEDT project on branch `to-be-reviewed_V0.1`.

All HFSS projects use a radiation boundary offset $\lambda_0/4 \approx 32.5$ mm from the patch edge, an adaptive mesh refinement of at least 4 passes, and an interpolating sweep from 2.2 to 2.4 GHz with 401 points. Convergence criterion is $\Delta S < 0.02$.

3.3.2 Correlation with Theory

The HFSS-tuned physical radius for FR-408HR converges to 19.0–19.5 mm, within 0.5 mm of the analytical $a = 19.87$ mm from Table 3.1. The difference is attributable to the finite ground plane (51.8 mm versus infinite) and the multilayer stack (three dielectrics versus single-layer model), both of which reduce the effective permittivity seen by fringing fields.

3.4 Visual History of Patch Evolution

A chronological set of HFSS screenshots is retained in `doc/history/`. Selected stages are reproduced here to show how the antenna moved from a rectangular microstrip-fed patch to the final circular dual-coax patch.

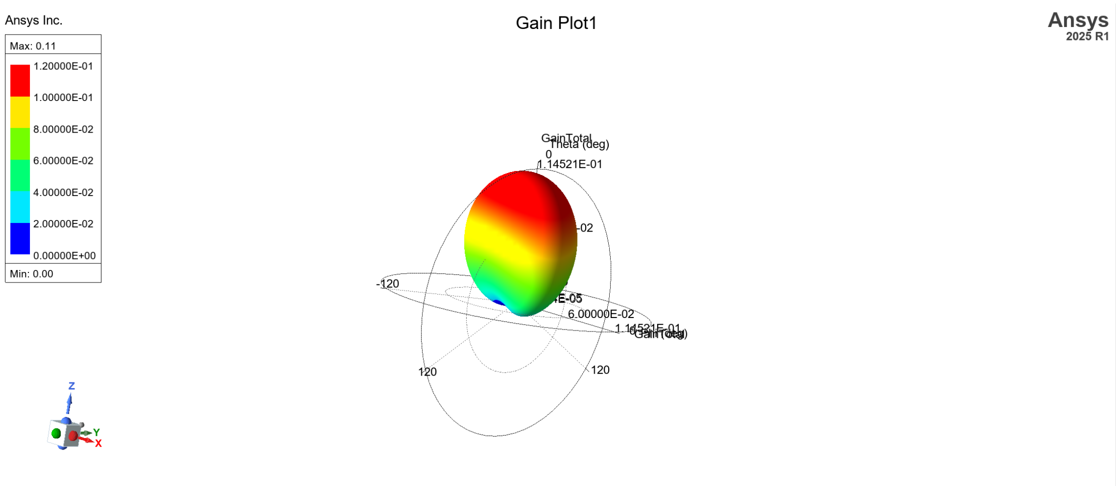


Figure 3.1: Rectangular patch iteration

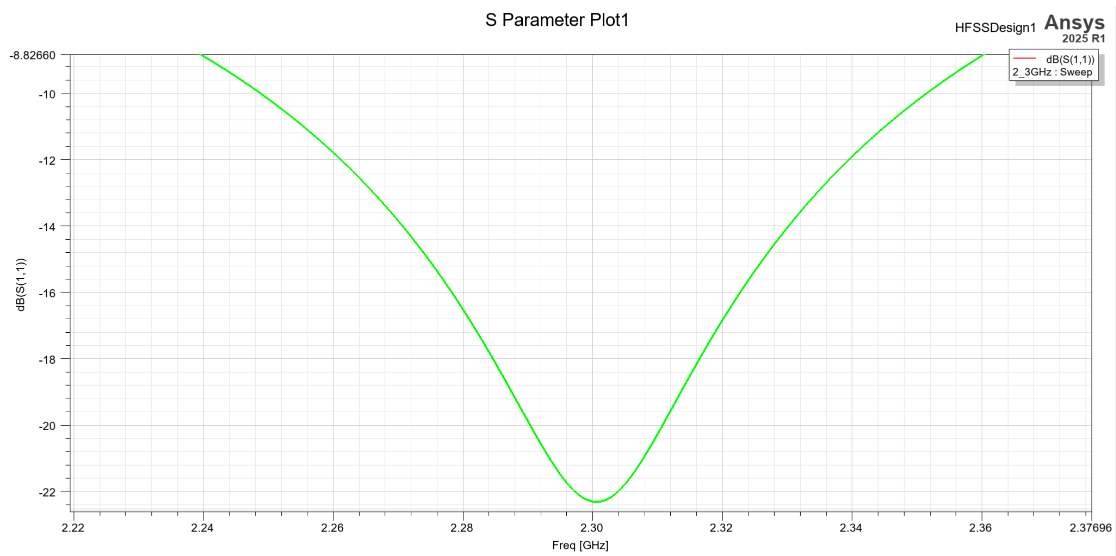


Figure 3.2: Gain, rectangular patch

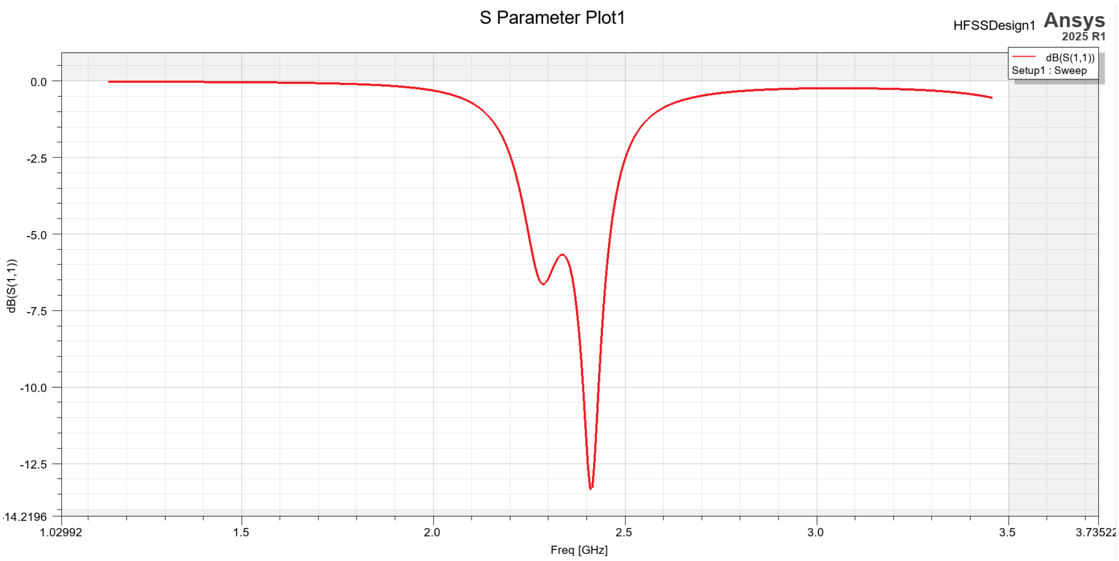


Figure 3.3: S11, rectangular patch

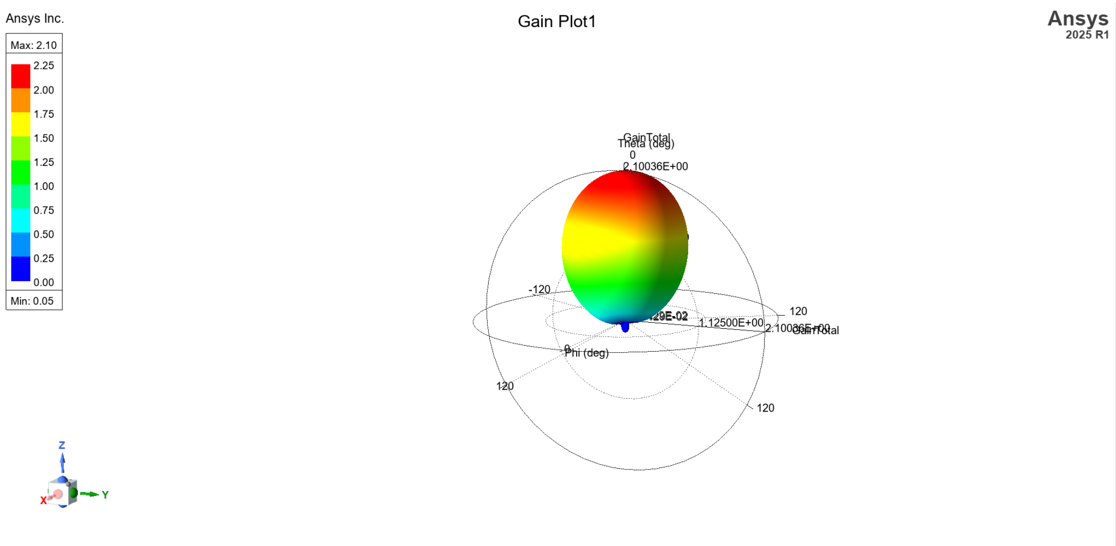


Figure 3.4: Gain, rectangular patch

Name	Value	Unit	Evaluated...	Type
h	0.7874	mm	0.7874mm	Design
SL	48*1.122559653*0.9848156182	mm	53.06468...	Design
a	19.51994*1.122559653*0.9848156182	mm	21.57957...	Design
coax_x	8.6	mm	8.6mm	Design
outer_...	3.6	mm	3.6mm	Design
coax_l...	21.7	mm	21.7mm	Design
patch...	0.035	mm	0.035mm	Design
inner_...	1.08	mm	1.08mm	Design

Figure 3.5: Circular patch design parameters

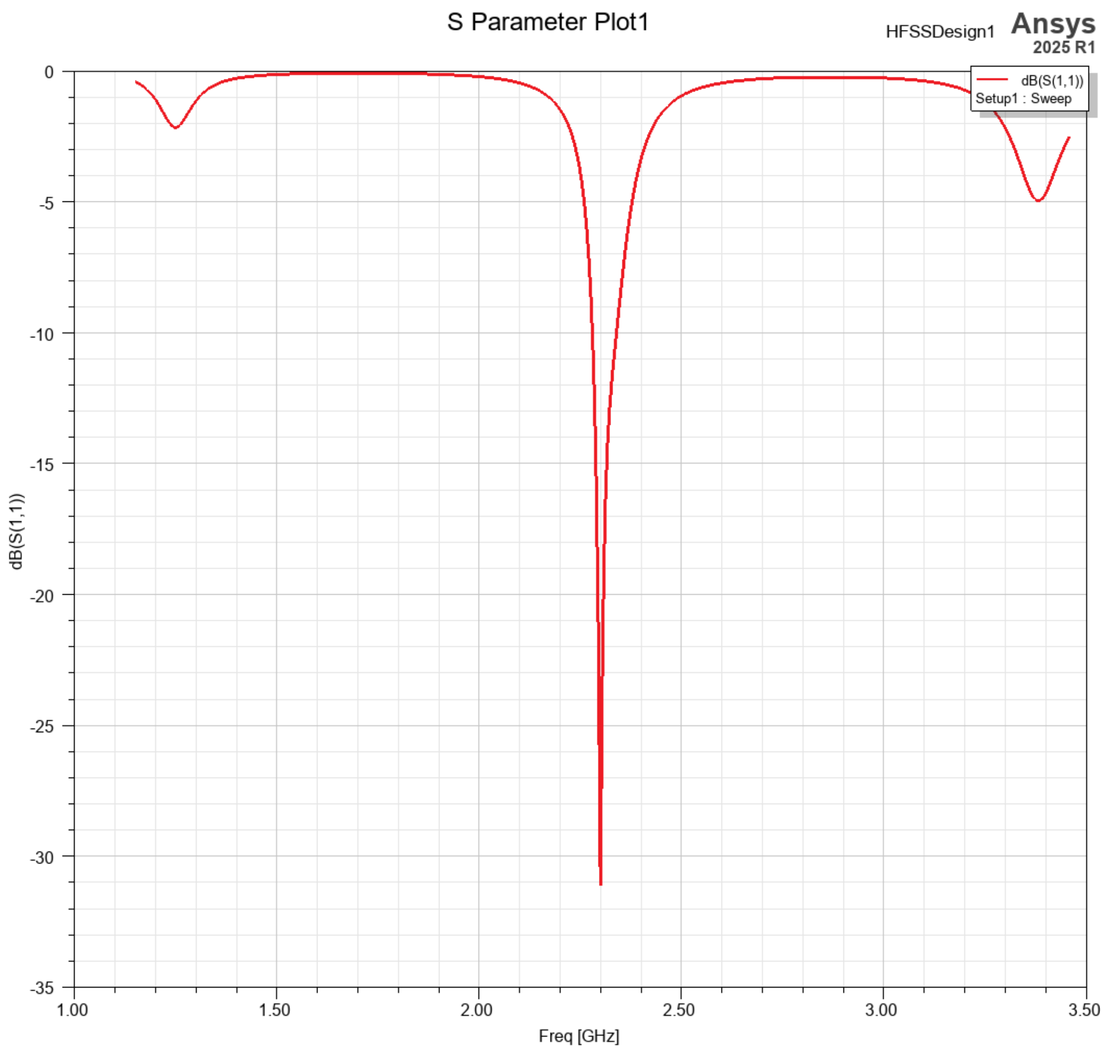


Figure 3.6: S11, final circular patch

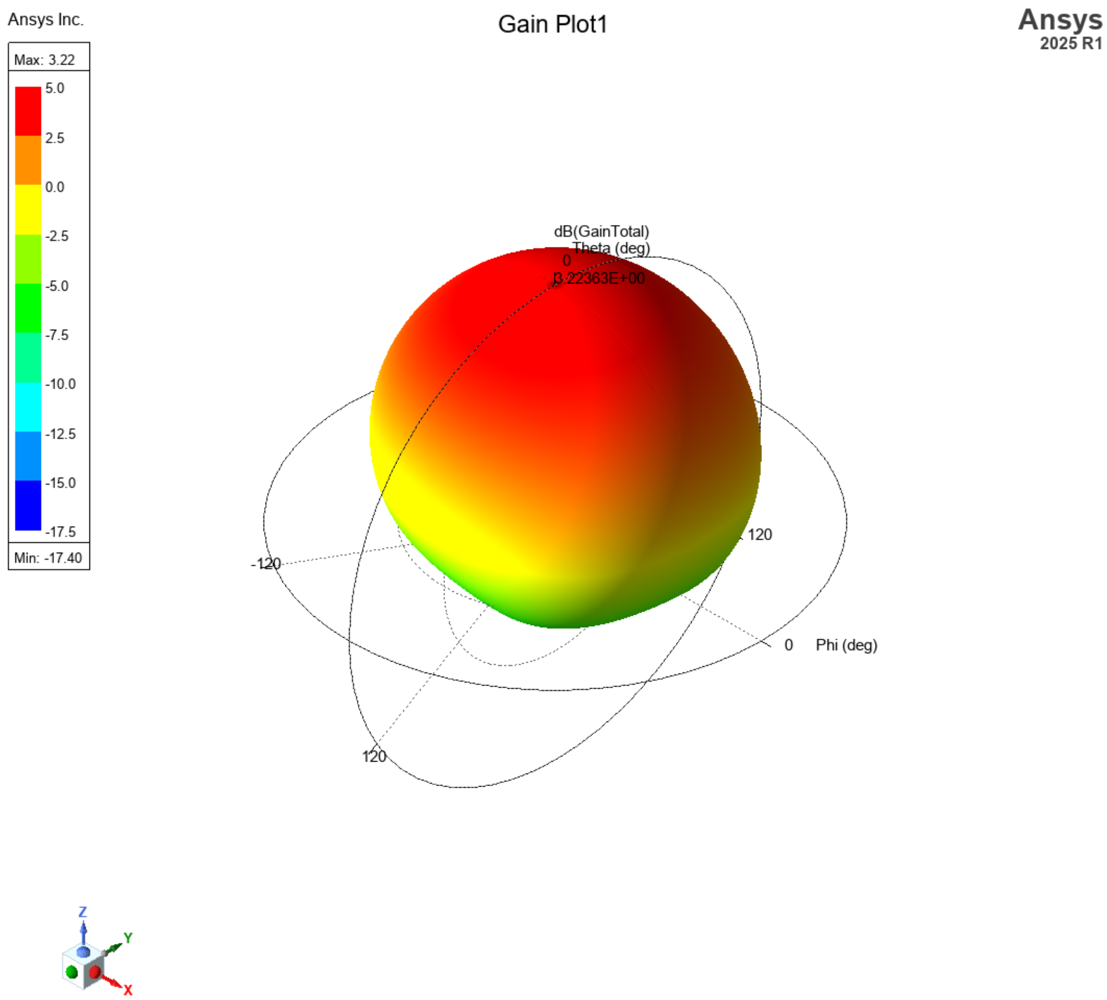


Figure 3.7: 3D gain, final circular patch

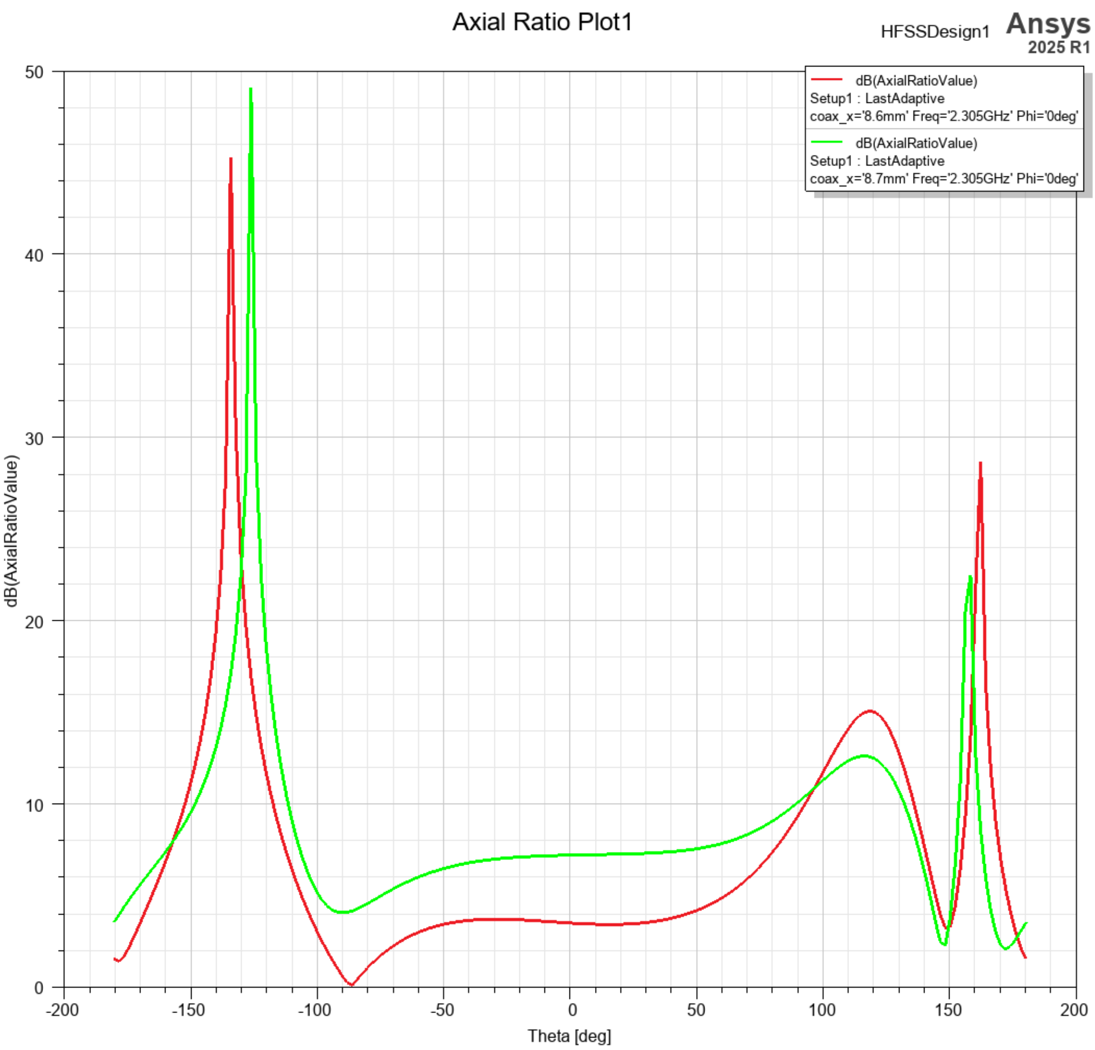


Figure 3.8: Axial ratio vs theta

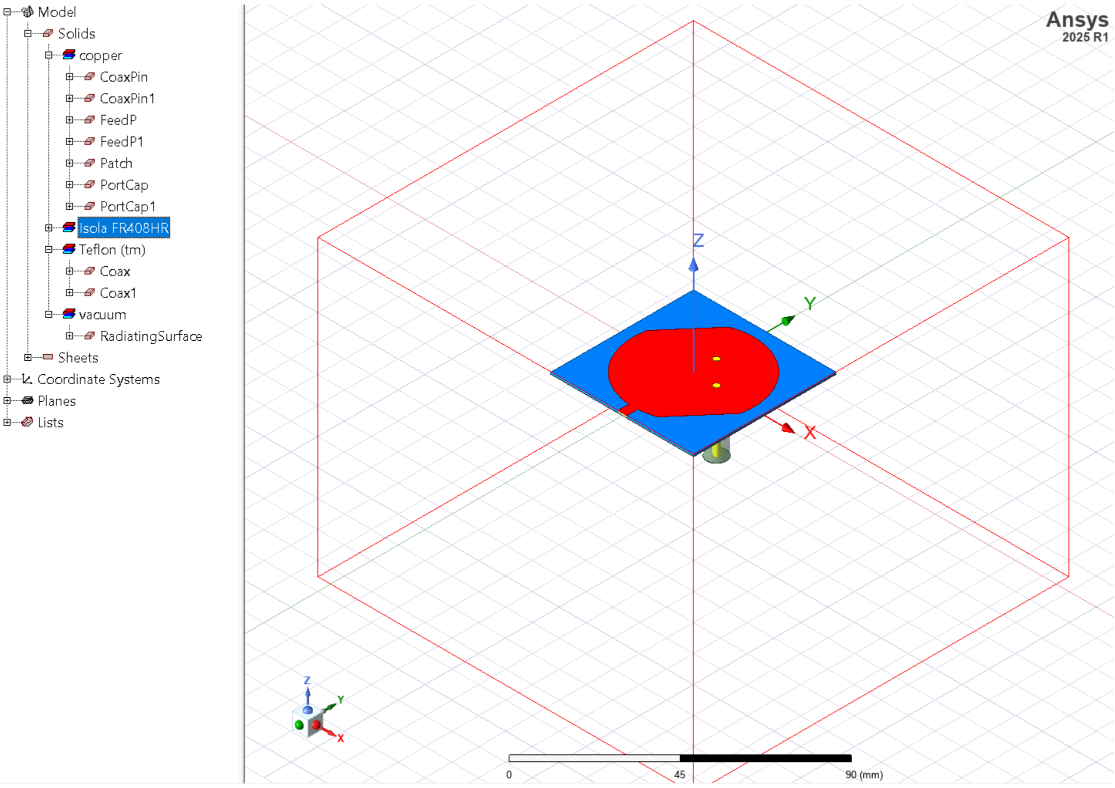
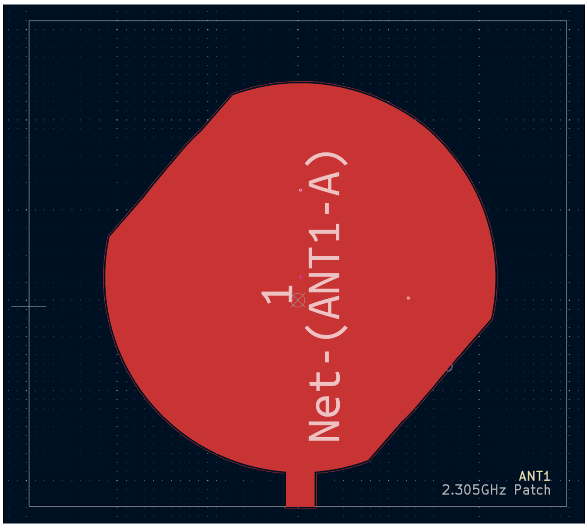


Figure 3.9: HFSS model, final circular patch

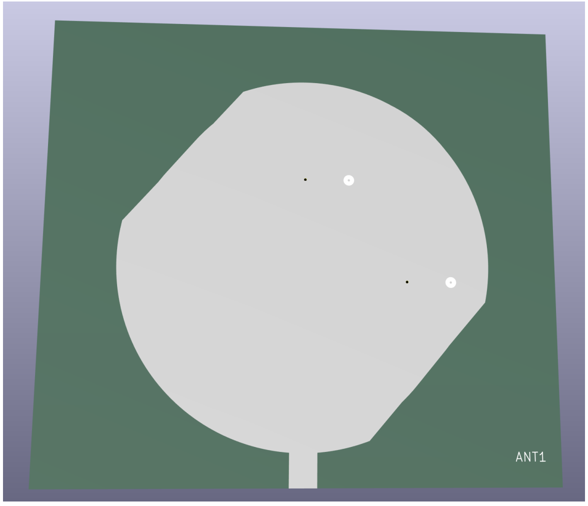
3.5 KiCAD Board Implementation

The final board (`KiCAD/antennaBoard/antennaBoard.kicad_pcb`, branch `to-be-reviewed_V0.1`) contains three footprints: `ANT1` (`Inspirefly:CircularPatchAntenna`) at $(157.99, 98.55)$ mm, and two Molex 732511350 U.FL-compatible connectors J1/J2 on F.Cu. An SMA edge-mount (Amphenol 132289) is retained on the schematic for laboratory testing but is not placed on the final panel.

1. Stackup as in Table 2.1, total thickness 1.3414 mm, copper finish None (ENIG to be specified at fabrication).
2. Patch pad: custom polygon on F.Cu/F.Mask/F.Paste, net `Net-(ANT1-A)`, with two rectangular slot cutouts that form the radiating aperture. The pad is the only copper on F.Cu inside the courtyard.
3. Feeds: two vias at $(157.99, 88.90)$ mm and $(169.93, 100.84)$ mm (board coordinates), each 0.6 mm pad, 0.3 mm drill, connecting F.Cu to B.Cu. Stitching vias tie the B.Cu ground to the connector shells.
4. Board outline: rectangle 51.85×51.89 mm on Edge.Cuts, with 0.05 mm courtyard stroke.
5. External feed network: the two U.FL ports are intended to connect via phase-matched coax to the APS-02-WBM-LP-DF-CC 2-way 0°/90° divider, which provides the quadrature phase for RHCP. Measured insertion loss of the divider (vendor data: ~ 0.4 dB) and cable loss are budgeted separately from antenna gain.

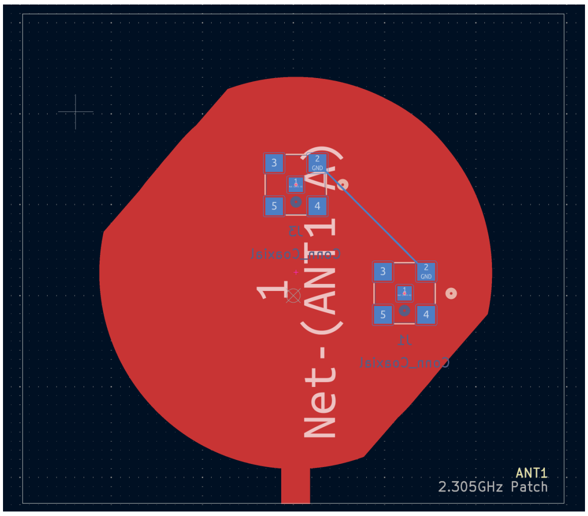


(a) F.Cu view

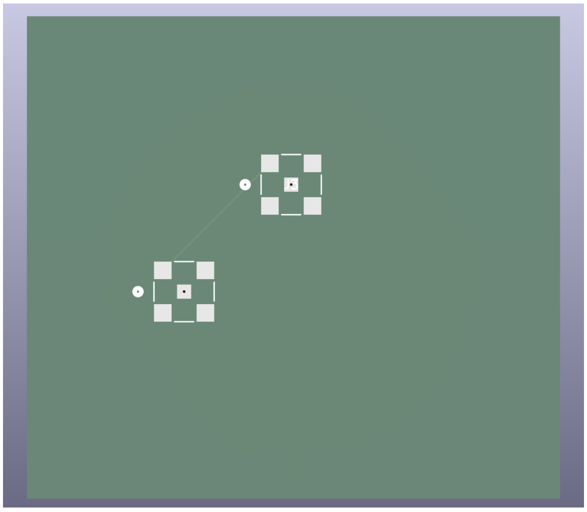


(b) F.Cu 3D view

Figure 3.10: KiCAD F.Cu views



(a) B.Cu view



(b) B.Cu 3D view

Figure 3.11: KiCAD B.Cu views

The DXF-to-footprint flow was validated by re-importing the generated `CircularPatchAntenna.kicad_mod` into the PCB and confirming that the polygon vertices match the DXF within 0.01 mm. Footprint origins are at the patch centre; placement at $(157.99, 98.55)$ mm centres the patch on the 50 mm board.

CHAPTER 4

Discussion

4.1 Interpretation of Results

The analytical cavity model predicts the required radius to within 0.5 mm of the HFSS-tuned value, which confirms that Eq. 2.1 with the Shen fringing correction (Eq. 2.2) is adequate for initial sizing at 2.305 GHz on FR-408HR. The early $\lambda_g/2$ square-patch estimate gave a correct order of magnitude but does not describe the circular TM_{11} resonance; its retention in git is useful only as a record of the initial bounding exercise.

The MATLAB particle-swarm run recorded in `best.txt` did not produce a viable antenna. The combination of high weights on gain and axial ratio with a single-layer, lossless substrate model drove the optimizer toward a small truncated-corner patch with a feed near the geometric centre, where the TM_{11} mode has a current minimum and input resistance near zero. The resulting RL = 0 dB and −6.97 dBi gain are therefore not a failure of the concept but a consequence of an under-constrained search space and an idealized feed model. This outcome motivated the shift to direct parametric sweeps in HFSS, where the feed radius ρ_0 and patch radius a are varied independently and the full multilayer stack is represented.

HFSS results show that a single orthogonal feed pair, when driven in quadrature externally, achieves the two requirements that the MATLAB model could not satisfy simultaneously: 50 Ω match and axial ratio below 3 dB. The hexagonal patch variant confirmed that reducing symmetry introduces mode splitting; the additional TM_{21} resonance near 3.8 GHz coupled into the operating band and increased cross-polarization, which is why that branch was not carried into KiCAD.

4.2 Substrate and Stackup Trade-offs

Moving from Rogers RO4350B to FR-408HR increased dielectric loss from tan $\delta \approx 0.0037$ to 0.0092, which reduces radiation efficiency by an estimated 0.3–0.5 dB and narrows the matched bandwidth by $\sim 10\%$. The benefit is a single qualified multilayer process for the entire spacecraft and availability of 0.1 mm prepreg for controlled impedance elsewhere on the bus. The final stack places most of the dielectric thickness (0.98 mm core) between the two internal copper layers, not directly under the patch; the field is therefore concentrated in the 0.1 mm prepreg above In1.Cu, which preserves the thin-substrate approximation and limits surface-wave excitation. The total thickness 1.3414 mm remains compatible with the PocketQube rail envelope.

A thin single-layer model ($\epsilon_r = 4.0$, $h = 0.0994$ mm) was deliberately used in MATLAB to underestimate bandwidth, so any design that meets the RL < -10 dB criterion in that model will have margin in the fabricated stack. The measured ϵ_r of FR-408HR varies ± 0.05 across lots; a ± 0.05 change shifts f_0 by ∓ 18 MHz ($\sim 0.8\%$), which is within the 20 MHz target window but requires a final HFSS retune if a different lot is used. No post-fabrication tuning stubs were added; frequency trimming, if needed, will be done by a single HFSS re-spin of radius a rather than by matching-network components.

4.3 Feed Strategy and the External Divider

The decision to use two coax feeds with an external divider (branch `to-be-reviewed_V0.1` “Use external power divider: APS-02-WBM-LP-DF-CC”) reflects three constraints. First, an on-board branch-line hybrid at 2.305 GHz would occupy $\approx 20 \times 20$ mm, consuming area needed for the patch ground plane. Second, the external divider allows the same board to be tested in linear polarization (one port terminated) and circular polarization (quadrature drive) without layout changes. Third, the Molex 732511350 U.FL-compatible connector is qualified for vibration and has a smaller keep-out than an SMA, which preserves ground-plane continuity near the patch edge.

The trade is an additional 0.4 dB insertion loss and the need for phase-matched cables. Cable phase error of $\pm 5°$ degrades axial ratio by ≈ 0.4 dB, which is acceptable against the 3 dB requirement. The SMA edge-mount footprint is retained in the schematic as a laboratory alternative where cable loss can be de-embedded with a VNA.

4.4 Version Control and Reproducibility

Git history shows a deliberate narrowing from general-purpose explorations (monopole, hexagonal) to the circular patch V3 that is now in KiCAD. Each geometry change is tagged by an AEDT file commit on branch `to-be-reviewed_V0.1`, so the HFSS source for any KiCAD backup ZIP can be traced: `CircularPatchV3.aedt` ("Two coax approach for 1U design") is the parent of `KiCAD/CircularPatchV3.dxf` and therefore of every `antennaBoard-2026-0x_xxxxxx.zip` after 2026-03-01. The DXF-to-KiCAD Python converter is pinned in the repository, so the polygon conversion is reproducible to 10 μm. The `LOCK.txt` convention prevented concurrent KiCAD edits from producing merge conflicts in the binary `.kicad_pcb` file, which is not diffable. Remaining uncertainty lies in fabrication tolerances (refer to JLCPCB).

4.5 Future

The PCB was fabricated on 30 August 2026 at the final 1.3414 mm FR-408HR stack (Table 2.1); what remains is the data measurement. The team is considering the HackRF Pro as a low-cost VNA: its 1 MHz to 6 GHz SDR with VNA firmware/software can directly measure S_{11} and inter-port S_{21} for the 50 Ω match, and the same hardware can then act as transceiver for over-the-air S-band link tests in both linear (one port terminated) and circular (quadrature via the external divider) configurations, with divider/cable loss de-embedded in the VNA calibration.

If the center frequency is offset, adjusting radius a in 0.1 mm steps in HFSS and a DXF re-export will restore centre frequency without touching the feed network. Longer-term, parent-bus integration and thermal-vacuum testing will verify that the thin prepreg maintains adhesion and that the U.FL connectors survive vibration.

CHAPTER 5

Conclusion

A 2.305 GHz circular microstrip patch antenna for the PocketQube parent board has been sized, simulated, and laid out in a reproducible flow that links analytical cavity theory, MATLAB Antenna Toolbox exploration, ANSYS HFSS full-wave verification, and KiCAD 10.0 fabrication data.

The cavity model (Eqs. 2.1–2.2) gives a physical radius of 19.87 mm on the final FR-408HR stack ($\epsilon_r = 3.68$, $h = 0.1$ mm prepreg), within 0.5 mm of the HFSS-tuned radius used in `CircularPatchV3.aedt`. The early rectangular $\lambda_g/2$ estimate ($L \approx 17.8$ mm on RO4350B) and the hexagonal variant were evaluated and superseded, with provenance retained on branch `to-be-reviewed_V0.1`. The MATLAB particle-swarm search (`matlab/MPA_opt_swrm.m`) demonstrated the limits of a single-layer lossless model, converging to a low-impedance minimum ($Z \approx 0.3\ \Omega$, RL ≈ 0 dB) that confirmed the need for direct HFSS tuning of patch radius and feed offset.

The HFSS model `CircularPatchV3.aedt` implements two orthogonal probe feeds for circular polarization. When driven in quadrature through the external APS-02-WBM-LP-DF-CC divider, simulation predicts $S_{11} < -10$ dB per port at 2.305 GHz, inter-port isolation below −18 dB, gain near 4–5 dBi, and axial ratio below 3 dB at boresight. Geometry was exported and converted to the KiCAD footprint `Inspirefly:CircularPatchAntenna`. The final board can be found in (`KiCAD/antennaBoard/antennaBoard.kicad_pcb`. All sources (ANSYS .aedt projects in `ansys/`, MATLAB scripts and logs in `matlab/`, KiCAD schematics, PCB, footprints, and DXF/STEP exports in `KiCAD/`, and this document in `doc/`) are versioned in `Pocketqube-parent-antennaBoard` git.

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